

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** of experience architecting distributed systems across social media, video streaming, and live commerce platforms. Owned end-to-end delivery of **YouTube** content monetization infrastructure serving 2 billion users, scaled **Google Search** indexing pipelines processing 100+ billion documents daily, and built **Whatnot** real-time bidding systems supporting millions in gross merchandise volume. Led teams of 8 engineers across time zones, partnering with product managers and data scientists to ship features used by tens of millions daily. Seeking to apply high-throughput systems and team leadership at a scale stage company.

Core Skills

Languages and Frameworks: JavaScript, SQL

Distributed Systems: Kubernetes, gRPC, Protocol Buffers, Apache Kafka, Cassandra, Redis

Backend: Java, Go, Python, C++, Spanner, Bigtable, MapReduce

Frontend: TypeScript, React, HTML/CSS, WebSocket

Infrastructure: GCP, AWS, Borg, Terraform, Prometheus, Grafana

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Established on-call rotation and runbook standards for 5 backend services, decreasing mean time to resolution for Sev1 incidents from 45 minutes to 12 minutes.
- Designed real-time notification service using **Go** and **Kafka** to deliver instant updates when video demonetization events occur, replacing batch jobs with sub-second latency.
- Architected content monetization pipeline processing creator earnings calculations across subscription and ad revenue streams for 2 billion monthly active users.
- Led migration of legacy PHP services to **TypeScript** microservices, reducing API response times by 40% for partner integration endpoints.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Designed and shipped seller analytics dashboard using **React** and **PostgreSQL**, reducing seller support tickets by 30% through transparent payout visibility.
- Built real-time bidding engine in **Go** handling 50,000 concurrent auction events during peak live shopping streams, supporting 7-figure daily GMV.
- Implemented event-driven architecture with **Kafka** to sync inventory across mobile and web clients, eliminating oversell incidents during high-traffic drops.
- Mentored 3 mid-level engineers on distributed systems design, conducting weekly 1:1s and architecture reviews for critical path features.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers responsible for **Google Search** indexing infrastructure, delivering indexing freshness improvements that surfaced new content 25% faster for search queries.
- Owned YouTube search quality initiative reducing zero-result rates by 15% through improved query understanding and index sharding strategies across 8 data centers.
- Redesigned crawl frontier scheduler in **C++** to prioritize high-traffic domains, increasing crawl efficiency by 35% without additional hardware spend.
- Championed adoption of **Starlark** configuration language for pipeline definitions, standardizing workflows across 12 teams and eliminating configuration drift incidents.
- Led technical hiring panel conducting 200+ interviews, hiring 15 engineers across L4-L6 levels with 90% retention over 3 years.

Software Engineer | Google | 02/2011 - 02/2015

- Implemented automated canary analysis framework using **Python** that detected regressions 4 hours earlier than manual monitoring, preventing 3 major search quality incidents.
- Developed **MapReduce** pipelines in **Java** processing 100+ billion documents daily to generate search ranking features for core ranking model training.
- Built **Bigtable**-backed serving system for real-time search suggest, achieving 99.99% availability with p99 latency under 50ms globally.
- Contributed to open-source **Borg** tooling, improving job scheduling fairness for batch workloads by 20%.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and implemented Windows kernel components for USB device driver framework, shipping in Windows 8 release with zero critical bugs reported post-GA.
- Collaborated with hardware OEMs including HP and Dell to debug and resolve device compatibility issues affecting millions of enterprise customers.
- Developed driver verification tools in **C++** using formal specification methods, reducing BSOD bugs in partner hardware certification by 40%.
- Mentored 2 intern engineers through successful conversion to full-time positions, owning technical project direction and code review quality.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented automated test generation tool for **.NET** framework APIs, increasing test coverage from 65% to 88% for covered modules.
- Documented driver development best practices adopted by 200+ internal engineers, reducing onboarding time for new team members.

Education

B.S. Honors Degrees

Virginia Tech | 2004 - 2007